

Problem C:

The FloorSet Challenge: Data-Driven SoC

Floorplanning

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you are encouraged to submit your questions directly through the Git repository.**

Q&A

Q1. The write-up mentions that any deviation in dimensions for fixed-shape and preplaced blocks is a hard-violation and should result in a score of 10, but then there is also a soft-constraint formula given to compute these violations, which are contradictory in nature. Could you please specify whether to consider this as a hard or soft constraint?

cost.py provided in the GitHub repo doesn't explicitly consider hard-constraint penalty ($M=10$) or the alpha, beta and gamma values given in the cost function. Does this mean that the organizers are yet to provide a proper validator program?

A1. Here is the contest repo, that addresses these concerns.

<https://github.com/IntelLabs/FloorSet/tree/main/iccad2026contest>

Q2. I am writing to request clarification regarding the technical environment and evaluation process for Problem C.

Specifically, I would like to inquire about the following details:

1. Memory Capacity: Could you please provide the specific memory (RAM/VRAM) limits allocated for each submission during evaluation?
2. Batch Processing: During the validation and test phases, can the test cases be processed in batches, or they should be evaluated sequentially?

A2. These are the hardware specs -> A100 80GB VRAM GPU, ICELAKE CPU with 48 cores, 128GB RAM

Batch processing -> The test cases will be evaluated sequentially.

Q3. I am writing to clarify the hardware specifications for the hidden test set evaluation environment, as this heavily impacts our inference strategy and the "Runtime Factor" score.

Could you please provide the following details regarding the evaluation server:

1. What specific CPU and GPU models will be used for evaluation?
2. What is the maximum RAM capacity allocated per submission?
3. Are we permitted to utilize multiprocessing or multithreading during inference?

A3.

1. A100 80GB VRAM GPU, ICELAKE CPU with 48 cores
2. 128GB RAM
3. The test samples are evaluated sequentially. But for each sample evaluation, multiprocessing or multithreading can be used.

Q4. We found that the training data golden answers violate MIB, Boundary, and Group constraints in some cases.

Since these are soft constraints, are the golden answers themselves penalized?

A4. Yes, seeing soft violations in training data is normal and consistent with how the dataset is intended to be used.